

From Proof Certificates to Mathematical Uptake

A Critical Reanalysis and Multiscale Research Program for AI-Generated Mathematics

Research synthesis building on Viteri and DeDeo (2022)

1 August 2026

Abstract

AI systems increasingly produce kernel-checked formal proofs, but correctness is only one property of a mathematical contribution. This report critically audits a large exploratory extension of [Viteri and DeDeo \(2022\)](#), recovers the exact source of an apparent power-law discrepancy, repairs paired analyses, and develops a research program connecting proof networks to cognitive science, philosophy of mathematical practice, and AI for mathematics. The original 2022 power-law pipeline fixed $x_{\min} = 10$; the newer pipeline selected $x_{\min}$ by Kolmogorov–Smirnov optimization. The estimates therefore measure different portions of the degree distribution. The defensible invariant is heavy-tailed reuse, not a universal pure power law or a cutoff-free exponent.

In 2,321 same-statement human/AI Lean-script pairs from 12 coarse source groups, AI scripts contain more local decomposition (median **have**: 3 versus 6; median paired difference +1) while strict and loose source-level premise heuristics have median paired difference zero. A new elaborated-term analysis retains 312 of 494 paired graphs after excluding either member if Lean error recovery introduced **sorryAx** or **SyntheticOpaque**. AI terms are modestly larger in the pair-weighted analysis (median paired +290 nodes), although equal-source uncertainty includes zero; they are slightly deeper and have a lower constant-node share. Their apparent heavier reuse tail at low cutoffs disappears at the publication convention $x_{\min} = 10$. These results support a narrow hypothesis: current AI proof production often adds internal scaffolding without clearly expanding the named library interface. They do not establish that AI uses less mathematical knowledge, produces epistemically inferior mathematics, or has a universal proof-network signature.

The central proposal is a four-scale framework separating (i) kernel-checkable derivation, (ii) human-facing proof presentation, (iii) library-level reuse and maintenance, and (iv) social uptake and trust. It yields preregisterable experiments in which proof content, presentation, authorship labels, and optimization objectives are independently manipulated.

1 Executive assessment

The prior work is ambitious and contains real discoveries, especially its willingness to falsify its own early claims. Its strongest contribution is not a single scalar measure of “AI mathematical style.” It is a set of contrasts that reveal why such a scalar is unlikely to exist. Proof scripts, elaborated proof terms, declaration graphs, generated certificate stores, and the dynamics of a human reader are different objects. Treating them as interchangeable made several early findings look stronger and more unified than the evidence allows.

After repair, four conclusions remain.

1. **The power-law discrepancy has an exact provenance.** The archived publication script set `cutf=10` and called `powerlaw.Fit(..., xmin=cutf)`. The later automatic fit commonly chose $x_{\min} \approx 1$. Neither estimate is intrinsically wrong, but they are not interchangeable.

2. **The paired source evidence is about decomposition, not library avoidance.** Same-statement AI scripts use more `have` statements. Once comments and strings are stripped, the claimed premise gap vanishes under both loose and strict regex definitions.
3. **Elaborated terms add a second, non-regex signal.** In the clean paired subset, AI proof terms tend to contain more internal nodes and edges and contain a smaller fraction of constant nodes. The size interval crosses zero after giving each source group equal weight, while the constant-share contrast remains directional. This is compatible with additional local scaffolding. It is not equivalent to lower mathematical sophistication or weaker understanding.
4. **Macro-level contrasts remain descriptive.** A generated certificate layer designed to be terminal will naturally have few later citations. Comparing it to a maintained human library confounds authorship with artifact role, chronology, and design objective. Such cases motivate hypotheses but do not identify an authorship effect.

The appropriate successor project should therefore ask: *which production objectives generate which proof artifacts, and how do those artifacts affect human validation, understanding, reuse, and repair?* This question is simultaneously computational, cognitive, and philosophical.

2 What was fixed, and what still required repair

Claude’s revision correctly withdrew several headline claims. In particular, it stopped interpreting a non-significant difference as equality, stripped comments and strings before counting premise-like tokens, narrowed a rank calculation to its selected block, and replaced overlapping one-way sums of squares with a nominally joint model. These are substantive improvements. The audit nevertheless found additional defects summarized in [table 1](#).

The most important meta-level correction is epistemic: a long exploratory pipeline should not end by selecting only analyses that fit one narrative. The report’s reversals are scientifically valuable, but they imply that future work needs frozen estimands, data-exclusion rules, and a distinction between confirmatory and exploratory stages.

3 Foundations: four objects, four scales

3.1 Why kernel correctness is not the whole object

A Lean kernel checks whether a proof term inhabits a type. This is an extraordinarily strong guarantee about formal derivability, conditional on the statement, definitions, axioms, elaborator, kernel, and computing stack. It does not by itself tell us whether the formal statement faithfully expresses the intended theorem, whether a human can see the proof’s main idea, whether its lemmas will be reused, or whether the artifact is easy to repair. Recent formal AI systems exploit precisely this grounded verification signal ([Yang et al., 2023](#); [Hubert et al., 2026](#); [Ren et al., 2025](#)); that makes these other questions more urgent, not less important.

The philosophy of mathematical practice has long resisted identifying ordinary proof with a bare formal derivation. [Rav \(1999\)](#) emphasizes that proofs carry methods and mathematical knowledge beyond their concluding propositions. Work on explanation distinguishes correctness from unification, dependence, and the revelation of a result’s characteristic structure ([Mancosu, 2008](#)). Rigour itself has been analyzed as plural and practice-sensitive ([Hamami, 2021](#); [Tanswell, 2024](#)), while fallibilist accounts locate justification partly in a community’s capacity to expose and

Table 1: Audit of the revised analysis. “Resolved here” means that the final synthesis either corrects the code or withdraws the affected inference.

Issue	Why it matters	Resolution in this report
Power-law cutoff	Automatic KS selection and fixed $x_{\min} = 10$ estimate different tails.	Recovered the archived production code; report a cutoff sweep and model-family caveat.
Premise regex	Capitalized or dotted source tokens are not authoritative elaborated dependencies.	Use regex results only as script-interface proxies; add elaborated constant-node measures.
“TOST-style” label	Percentile-CI containment inside a post hoc 10% margin is not a formal TOST.	Call it a descriptive smallest-effect-size-of-interest check; do not claim established equivalence.
Variance bootstrap	Each row, rather than each resampled theorem instance, received a new pseudo-problem label, producing zero-width CIs.	Correct the relabeling; for complete blocks use an exact theorem/system/residual sum-of-squares partition on $\log(1 + y)$.
HF complete-block search	Forty random triples among 45 systems missed large valid blocks.	Exhaustively seed all triples; recover a 138×4 complete block.
Stale report sections	Corrected results coexisted with old 7-system ranks and overlapping 89.8%+16.4% variance claims.	Treat the old Markdown as an exploratory lab notebook, not a citable final report.
Lean error recovery	Many extracted graphs contained sorryAx/SyntheticOpaque ; these are not completed proofs.	Exclude a pair if either graph contains either label; disclose 312/494 retention.
Term truncation flag	term0 uses zero expansion levels, leaving a constant frontier and marking every graph “truncated.”	Interpret term0 as the unexpanded root proof-term DAG; ignore that flag as an attrition indicator.
Belief simulations	The expensive paired run ended at 125/157 and wrote no final paired result; graph means also mix isolates with connected nodes.	Do not use the incomplete paired simulation as evidence; preregister node conditioning and stopping rules in future work.
ETP provenance	A shortest Dijkstra path is one arbitrary certificate among possible paths; generated outputs are terminal by design.	Downgrade causal language and propose randomized retention/organization experiments.

correct errors (De Toffoli, 2021). Formalization changes this ecology, but does not erase it (Avigad, 2026; Avigad et al., 2024; De Toffoli and Tanswell, 2026).

3.2 The proposed multiscale map

The 2022 epistemic-phase-transition (EPT) model concerns a hypothetical reader’s changing beliefs over a dependency network (Viteri and DeDeo, 2022). It is best interpreted as a model of reception or justification under local uncertainty. It is not an objective quality function for machine corpora. A verifier can collapse one kind of uncertainty while leaving statement correspondence, explanatory understanding, provenance, and social responsibility unresolved. Empirical studies support this broader view: mathematicians validate using formal, informal, and example-based reasoning (Weber, 2008); experts attend more to logical structure and implicit warrants than novices (Inglis and Alcock, 2012; Panse et al., 2018); and mathematicians often read published proofs for insight rather than line-by-line correctness (Mejía-Ramos and Weber, 2014). Interactive evaluations of language models

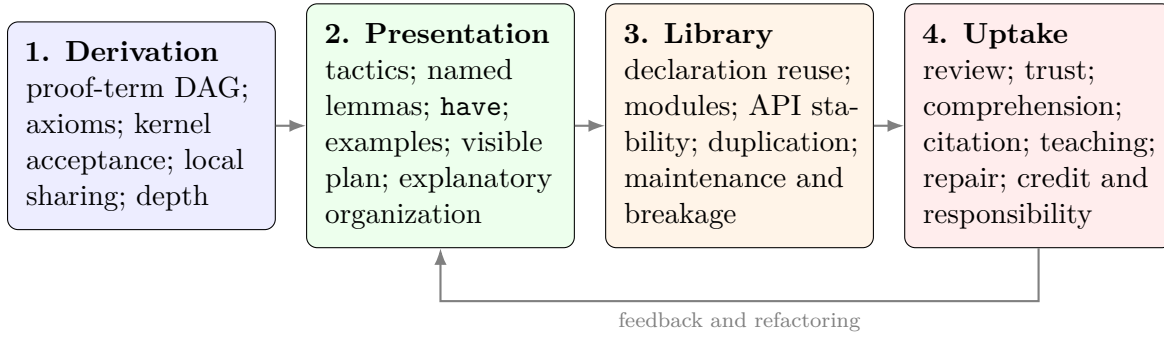


Figure 1: A proof artifact moves through rendering, integration, and use across distinct scales; later uptake can prompt feedback and refactoring. Arrows are empirical processes, not logical equivalences. The same kernel term can be presented in different ways, and the same presentation can receive different social treatment.

likewise find that users ask why a step was taken, not only whether the final answer is right (Collins et al., 2024).

4 The power-law discrepancy, resolved

The archive at `/Users/simon/Desktop/OLDER_RESEARCH_ARCHIVE/SCOTT` contains the script that fed the reported table. In `pl.py`, the operative lines are:

```
cutf=10
fit=powerlaw.Fit(data[0][1:], discrete=True,
                 xmax=data[0][0], xmin=cutf)
```

The production file `convert_final_check.rb` writes the node count and nonzero reuse degrees to `TEMP/data.dat`, invokes `./pl.py`, and prints the resulting exponent. The independent archive file `proof_dist.rb` also begins with `xmin=10`. This is direct provenance, not an inference from approximate numerical matching.

The later Python analysis used the standard `powerlaw` behavior of choosing $x_{\min}$ to minimize a KS distance. It often chose a cutoff near one. Fixed $x_{\min} = 10$ reproduces 40 of 47 archived exponents within 0.010; automatic estimates have essentially no cross-network correlation with the published numbers. This behavior is unsurprising. Tail inference depends strongly on where the tail begins, and model checking requires goodness-of-fit and comparisons to alternatives (Clauset et al., 2009; Alstott et al., 2014). In the recovered networks, a power law consistently beats an exponential, but a lognormal is frequently competitive. Strong scale-free claims therefore exceed the evidence (Broido and Clauset, 2019).

The revised conclusion is:

Proof dependency networks exhibit heavy-tailed reuse. A fixed-cutoff index near two summarizes the degree tail under the publication convention, but no cutoff-free or universally pure power-law mechanism has been established.

This also changes how authorship comparisons should be interpreted. An exponent difference at $x_{\min} = 2$ that disappears at $x_{\min} = 10$ is evidence of different moderate-degree structure, not a robust difference in the far reuse tail.

5 Data and confirmatory estimands

5.1 Same-statement proof scripts

The corrected NuminaMath analysis contains 2,321 pairs for which both a human and prover proof were marked valid and both yielded at least one premise-like token under the extraction rule. Pairs fall into 12 coarse source groups. Metrics are line count, tactic count, **have** count, and two source-token proxies for premise contact: a loose set of dotted or capitalized identifiers and a strict set of dotted identifiers after comments and strings are removed. These are presentation metrics; neither token count is a semantic premise census.

The primary descriptive estimand is the median within-pair difference. Uncertainty is reported both by resampling pairs and by resampling source groups. Because the latter estimand can still be dominated by a large source, the final output also reports an equal-source summary: calculate each source’s median paired difference, then take the median across sources. With only 12 source groups, all cluster-level inference is necessarily coarse.

5.2 Same-statement elaborated proof terms

For a 500-pair extraction sample, both human and AI **term0** graphs existed for 494 statements. The graph is the DAG of the theorem type and elaborated root proof term, with structurally identical expressions deduplicated and edges directed from a child premise to the containing expression. It does not recursively expand referenced theorem definitions. A pair is excluded if either graph contains a label matching **sorry** or **SyntheticOpaque**; 312 pairs remain. This 36.8% pairwise exclusion is substantial, and the retained sample must not be treated as a random draw from all valid proofs.

Metrics include nodes, edges, longest DAG path, expression-sharing ratio, maximum out-degree, out-degree Gini coefficient, constant-node count and share, and fixed-cutoff discrete tail-index approximations. Node and edge counts include both the theorem’s type and value under a virtual root; they are structural sizes, not human proof lengths. Constant nodes are elaborated references, but their number can still depend on elaboration and tactic expansion.

5.3 Matched theorem-by-system blocks

For the lean-eval records, ranking is performed only after selecting a complete 27×3 theorem-by-system block. For the HuggingFace census, exhaustive triple seeding and greedy enlargement recover a 138×4 block that the earlier random search missed. Friedman tests and Kendall’s W quantify stable rank ordering within each block. A two-way sum-of-squares partition on $\log(1 + y)$ attributes descriptive variation to theorem, system, and residual/interaction; these shares sum exactly to one. They do not estimate a universal causal authorship effect because system identity, benchmark, decoding, and data lineage remain entangled.

6 Results

6.1 Source scripts: more local decomposition, no surviving premise gap

The line-count difference is small relative to the distribution, but calling the proofs “the same length” requires a justified equivalence margin. The previous $\pm 10\%$ margin was generated from the human median after the fact and its CI-containment procedure was not formal TOST. The responsible conclusion is simply that the typical difference is a few lines in this corpus.

Table 2: Corrected same-statement source results. Intervals resample coarse source groups while retaining their observed sizes. P-values from thousands of pairs are not treated as measures of substantive importance.

Metric	Human median	AI median	Median paired Δ	Source-cluster 95% interval
Lines	23	26	+2	[1, 3]
Tactic invocations	12	11	0	see machine-readable supplement
have steps	3	6	+1	[0, 1]
Loose premise proxy	5	4	0	[0, 0]
Strict dotted proxy	3	2	0	[0, 0]

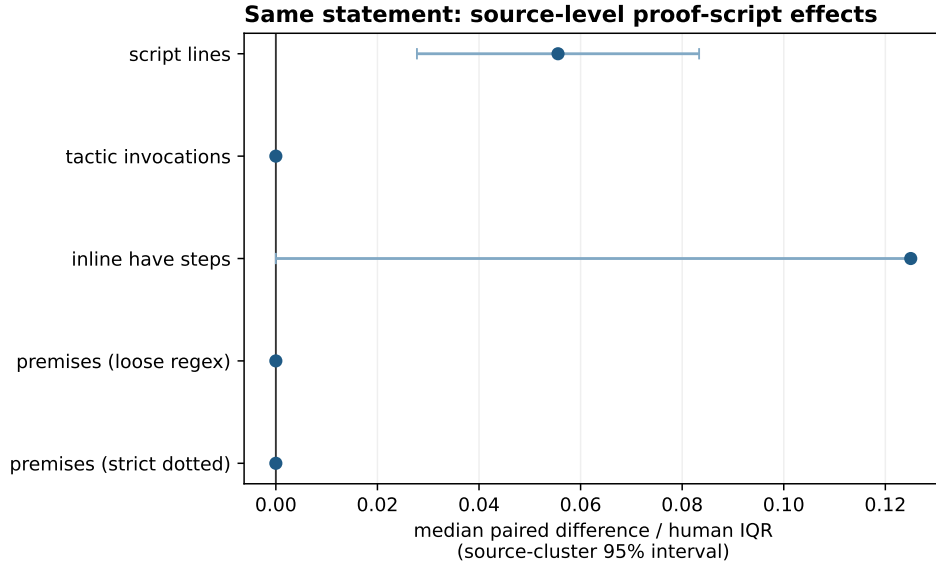


Figure 2: Paired source effects standardized by the human interquartile range. A zero-width interval can arise for discrete medians and 12 coarse clusters; it should not be read as literal certainty or as a formal equivalence result.

The **have** result is the most interesting surviving presentation contrast: the AI median is twice the human median, even though the median paired difference is only one because the distribution is zero-inflated and skewed. Giving every source equal weight yields a median difference of 0.5 and a bootstrap interval $[0, 1.5]$, so the result is a tendency rather than a universal signature. In contrast, the original claim that AI touched roughly half as much of the library was a measurement artifact. Removing comments and strings eliminates the paired gap under both token rules. The data therefore support finer visible decomposition, not reduced named-premise contact.

6.2 Elaborated terms: more internal structure, similar far-tail exponent

The pair-weighted conjunction of source and term results suggests an *internalization* pattern: similar visible contact with named premises, but more generated expression structure between those interfaces and the theorem. This pattern could arise from subgoal decomposition, explicit coercions, normalization, repeated tactic templates, or different choices of theorem APIs. It does not establish which mechanism is operative. Source-balanced intervals for nodes $[-6.5, 540.75]$ and

Table 3: Clean paired `term0` results ($n = 312$). The listed intervals are source-cluster bootstrap intervals from the audited run; all metrics are exploratory and correlated.

Metric	Human median	AI median	Median paired Δ	Source-cluster 95% interval
Nodes	3,816.5	4,561	+290	[20, 512]
Edges	7,145.5	8,632.5	+538.5	[39, 1,039]
Sharing ratio	1.875	1.895	+0.0089	[0.0047, 0.0121]
DAG depth	65	68	+1	[0, 3]
Maximum reuse	197.5	207.5	+2	[0, 10]
Reuse Gini	.4333	.4398	+.0030	[.0012, .0052]
Constant-node share	.0530	.0456	−.0046	[−.0061, −.0028]
α , $x_{\min} = 2$	2.102	2.057	−.024	[−.033, −.014]
α , $x_{\min} = 5$	2.330	2.303	−.028	[−.031, −.011]
α , $x_{\min} = 10$	2.531	2.527	−.009	[−.019, .013]

edges $[-8, 1025]$ cross zero, whereas the source-balanced constant-share interval $[-.00624, -.00071]$ does not. Thus proportional internal structure is more stable than absolute size across the 12 coarse source groups. A decisive follow-up should compare multiple proofs elaborated under the same Lean/Mathlib version and decompose node counts by expression type, tactic origin, and referenced declaration.

Most importantly, the low-cutoff exponent effect is not stable. At $x_{\min} = 10$, the paired median difference is near zero and its interval crosses zero. This aligns the paired AI comparison with the recovered publication convention and prevents a narrative about universally heavier AI reuse tails.

6.3 Matched systems: theorem effects are large, but style stability is dataset-specific

The corrected deterministic lean-eval block gives Kendall $W = 0.374$ for vocabulary ratio, 0.457 for lines, 0.652 for `have`, and 0.586 for distinct-premise rank. Thus the original 7-system rank table was invalid, but rank stability does survive in a narrower 3-system block. The recovered HuggingFace block gives $W = 0.534, 0.050, 0.121$, and 0.739 for the same four metrics, respectively. Line and `have` stability do *not* replicate, while premise and vocabulary ordering are strong in this selected block.

In the HF block’s exact $\log(1 + y)$ partition, theorem accounts for 86.4% of line-count variation, system for 0.8%, and interaction/residual for 12.8%. Distinct-premise count reverses that balance: theorem accounts for 33.2%, system for 56.6%, and interaction/residual for 10.2%. The question “problem or prover?” therefore has a metric-specific answer even inside one block.

This reversal is informative. “Verbosity is a stable trait of a prover” is not a corpus-free law. The HF block contains related dataset/model variants, and its result may reflect lineage or decoding configuration as much as base-model identity. The complete-block variance decomposition in the supplement is therefore descriptive. A credible variance-components claim requires a crossed generation design in which every theorem is sampled repeatedly from every independently defined system under a fixed toolchain and budget.

6.4 Why the macro-corpus results remain hypotheses

The Equational Theories Project (ETP) generated layer is an extreme and useful case: most generated declarations are never cited, and the layer has low modularity when viewed alone. But

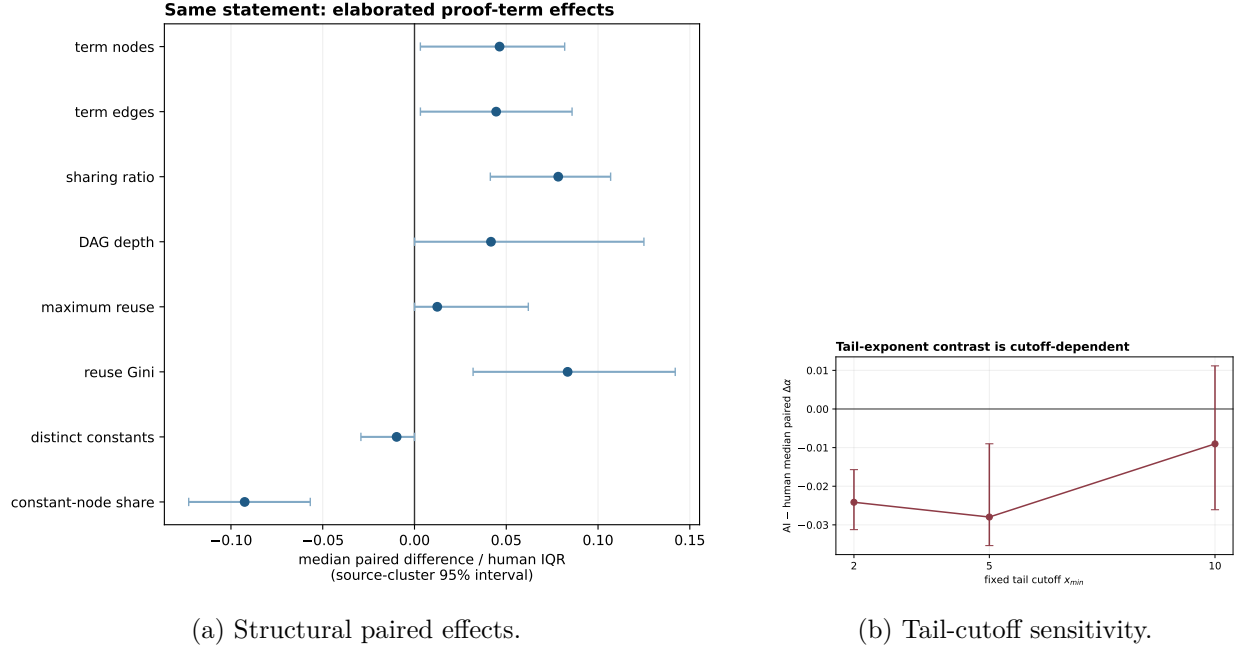


Figure 3: Elaborated proof-term comparisons. AI terms are larger and contain proportionally fewer constant nodes. The exponent contrast is present only when moderate-degree nodes enter the fit.

the artifact is a certificate store built to settle implications, not a long-lived API designed for future authors. Its terminality is partly an objective, not necessarily a machine cognitive limitation. Likewise, a Dijkstra-selected shortest provenance path does not identify “the” derivation when many derivations are possible.

The belief-dynamics analyses raise a separate identification problem. Mean belief over all nodes is highly sensitive to isolates and graph boundary choices. Comparing an explicit human layer to a generated terminal layer without conditioning on connected status can turn data organization into an apparent epistemic effect. The unfinished paired belief run—terminated after 125 of 157 pairs without a final result file—cannot resolve this. The macro results should be retained as case studies that motivate controlled interventions, not as evidence that machine mathematics is intrinsically fragile, heap-like, or non-modular.

7 Interpretation across cognitive science, philosophy, and AI

7.1 Cognitive science: model the reader, not only the artifact

The EPT model offers a generative bridge from dependency structure and local uncertainty to global confidence. Its next test should collect belief trajectories from readers. Structural metrics alone cannot validate cognitive plausibility. Expert proof reading involves backward looks, implicit-warrant reconstruction, examples, prior authority, and goal-dependent allocation of attention (Weber, 2008; Inglis and Alcock, 2012; Mejía-Ramos and Weber, 2014; Panse et al., 2018). These processes suggest measurable extensions: node-specific priors, selective attention, chunking into lemmas, trust in authors or verifiers, and stopping when expected checking value falls below its cost.

AI proofs are especially useful experimental stimuli because one kernel term can be transformed into several presentations while correctness is held fixed. A proof can be expanded or compressed,

renamed, reorganized into lemmas, supplied with examples, or labeled human/AI. This enables causal tests that historical proof networks cannot provide.

7.2 Philosophy: correspondence, explanation, and repair

Formal verification relocates rather than eliminates epistemic labor. If the term checks, attention moves to statement correspondence, definitions, axioms, imported libraries, and the provenance of the formalization. This is a correspondence problem between intended and formal mathematics. It also leaves explanation open: a proof may certify without displaying why a theorem depends on a characteristic structure or how its method unifies other results (Mancosu, 2008; Rav, 1999).

A useful operationalization of mathematical justification is *repair capacity*. Can another agent find the key dependency, locate a gap in an informal rendering, update the proof after an API change, and explain the change? This connects fallibilist correction (De Toffoli, 2021) to software-like maintenance without reducing mathematics to software engineering. It also avoids the false choice between perfect formal certainty and merely social acceptance.

7.3 AI for mathematics: optimize beyond pass rate

Current formal systems demonstrate that verified search can reach Olympiad-level performance (Trinh et al., 2024; Hubert et al., 2026), while retrieval and subgoal decomposition make premise access and proof organization explicit design problems (Yang et al., 2023; Ren et al., 2025). AI has also assisted human discovery by exposing patterns that mathematicians turn into concepts and proofs (Davies et al., 2021). These successes suggest a transition from proof scarcity to proof abundance.

If many valid certificates can be generated, the bottleneck becomes selection and presentation: which proof should enter a library, which lemmas deserve names, what should a reviewer inspect, and what artifact will teach or support future work? Training only for kernel acceptance or short tactic traces can produce locally efficient but socially expensive proofs. A research program should make those downstream costs visible to the objective.

8 A significant research program

8.1 Work package 0: measurement validity and an open proof observatory

Build a versioned extraction layer that records, for each generated proof:

- exact theorem statement, imports, Lean/Mathlib commit, options, axioms, and validation log;
- source syntax trees rather than regex tokens;
- elaborated proof terms with node type, constant identity, universe normalization policy, tactic provenance where available, and explicit error-recovery flags;
- declaration-level dependencies at the commit where the proof was produced;
- generator identity, prompt/search budget, samples attempted, wall-clock compute, and selection rule;
- links among informal problem, formal statement, source script, proof term, and later revisions.

Hand-audit a stratified sample and report precision/recall for premise extraction. Unit tests should include nested comments, quoted identifiers, namespaces opened locally, generated declarations, multiple proof bodies, and intentional failure cases. Publish hashes and manifests so a derived table can be traced to immutable inputs.

8.2 Work package 1: a genuinely crossed proof-production experiment

Select 150–300 theorems stratified by domain, statement length, library depth, and difficulty. For each theorem collect at least two independent human formalizations where feasible and three independent samples from each of four contrasting AI architectures: whole-proof generation, retrieval-augmented generation, interactive/agent search, and RL-guided formal search. Fix the Lean/Mathlib version and expose the same accessible premises and budgets.

The primary model should be preregistered before generation:

$$g(y_{t,s,r}) = \mu + A_s + T_t + (A \times D)_{s,d(t)} + u_{t,s} + \epsilon_{t,s,r}, \quad (1)$$

where theorem t , system s , repeat r , and domain d are crossed; g is specified in advance (for counts, usually $\log(1 + y)$ or a negative-binomial link). Report theorem, system, interaction, and repeat variance. Hold out theorem families, not random near-duplicate statements.

Primary outcomes should be few and mechanistic: internal expression nodes per distinct constant, named intermediate claims, proof-term depth, retriever recall, reuse concentration, and generated lemma survival. Power-law exponents are secondary sensitivity summaries with fixed cutoffs and alternative-distribution tests.

8.3 Work package 2: causal experiments on proof reception

Use kernel-equivalent presentations in a factorial reader study:

Factor	Levels
Presentation	compressed / expanded / hierarchically chunked
Provenance label	human / AI / unlabeled (randomized, sometimes deceptive with debrief)
Verifier information	none / “kernel checked” / audit trail and assumptions
Error status	valid / planted local gap / statement-correspondence mismatch
Reader	advanced undergraduate / graduate / domain expert

Measure confidence after each proof segment, validation accuracy, time, eye movements or navigation, recall of the main idea, transfer to a nearby problem, calibration, and willingness to reuse a lemma. Fit the EPT model to individual trajectories and compare it against sequential-evidence and attention/chunking alternatives. The key test is out-of-sample prediction: parameters inferred from one set of proofs must predict confidence and error detection on held-out proofs.

8.4 Work package 3: library uptake and maintenance

Follow accepted human-, AI-, and mixed-authorship pull requests longitudinally. Match or weight by domain, project maturity, declaration type, size, review policy, and chronology. Outcomes at 3, 6, and 12 months should include citations by later declarations, number of downstream files, review rounds, issue reports, API-breaking edits, proof repair time after dependency changes, and whether a lemma is generalized, renamed, deprecated, or taught.

The strongest design is randomized assistance within a real library: route eligible contributions to (i) ordinary generation, (ii) generation plus a library-architect agent optimizing reuse/API quality,

or (iii) generation plus a human editor. All outputs remain human-reviewed. This tests whether apparent “terminality” is caused by machines, objectives, or editorial organization.

8.5 Work package 4: an ecology of proof objectives

Train or rerank proof agents under controlled multi-objective rewards:

$$R = R_{\text{valid}} - \lambda_1 C_{\text{compute}} - \lambda_2 C_{\text{repair}} + \lambda_3 U_{\text{future}} \\ + \lambda_4 H_{\text{reader}} + \lambda_5 P_{\text{provenance}},$$

where future reuse U , human reception H , and provenance quality P are measured on held-out tasks rather than guessed from static proxies. Compare Pareto fronts rather than collapsing all values into one score. A proof can be short but opaque, readable but redundant, or reusable but expensive; those tradeoffs should remain visible.

8.6 Work package 5: dialectical and adversarial mathematical agents

Correctness filters do not guarantee that a formal statement is the intended one. Pair a proposer with independent agents assigned to statement auditing, counterexample search, dependency minimization, proof explanation, and hostile review. Preserve failed conjectures and objections as first-class research objects, following the dialectical tradition in mathematical practice (Corneli et al., 2017). Evaluate whether heterogeneous audit roles catch correspondence errors and improve human calibration more efficiently than simply sampling more proofs from one prover.

8.7 Preregistered hypotheses and falsifiers

Table 4: A compact confirmatory agenda.

ID	Prediction	Decisive falsifier or boundary condition
P1	At fixed theorem, toolchain, and accessible library, AI proofs have more internal expression nodes per distinct referenced constant.	The system effect vanishes after tactic family and repeat are crossed, or reverses in at least two independent architectures.
P2	Increased have use mediates part, but not all, of the term-size difference.	A controlled style intervention changes have count without changing term structure, or mediation fails on held-out systems.
P3	Chunking proof terms into meaningful named lemmas improves reader calibration and transfer more than equal-length arbitrary chunking.	Arbitrary and meaningful chunking perform equivalently on held-out proofs and experts.
P4	A kernel-checked label reduces line-by-line checking but does not eliminate statement-audit effort.	Verifier information changes neither navigation nor error detection, or suppresses both local and correspondence-error detection equally.
P5	Library-architect optimization increases later reuse and lowers repair cost without reducing acceptance.	Randomized architecture assistance yields no improvement at 6–12 months or merely inflates trivial citations.

ID	Prediction	Decisive falsifier or boundary condition
P6	Tail-exponent authorship differences are concentrated below $x_{\min} = 10$ and are not stable across toolchains.	A preregistered fixed- $x_{\min} = 10$ contrast replicates across domains, systems, and toolchain versions with alternative families rejected.
P7	EPT parameters predict segment-by-segment confidence better after adding attention and chunking than from graph structure alone.	A simple length/position baseline matches held-out predictive performance, or fitted parameters fail to transport between proofs.

9 Standards for claims about AI-generated mathematics

Future papers should state which of the following has actually been demonstrated:

1. **Formal validity:** a kernel checks a term under enumerated assumptions.
2. **Correspondence:** experts agree that the formal statement and definitions match the intended mathematical claim.
3. **Explanatory value:** readers identify why the result holds and transfer the method.
4. **Library value:** the contribution forms a stable, reusable interface.
5. **Social robustness:** provenance, review, responsibility, and correction mechanisms support warranted communal uptake.

No single network statistic certifies all five. The EPT model is most naturally linked to the fifth through a model of readers, while proof terms strongly address the first. The research opportunity lies in measuring the transformations between them.

10 Conclusion

The corrected analyses do not reveal a universal architecture of machine mathematics. They reveal something more actionable: current AI and human proofs of the same statement can expose a similar named-library interface while differing in internal scaffolding and presentation. The difference is modest, cutoff-sensitive, and strongly conditioned by corpus and extraction choices. At library scale, much larger contrasts often reflect what an artifact was built to do.

Viteri and DeDeo’s central question—how finite, fallible agents come to trust long deductive arguments—becomes sharper in the age of AI. Kernel verification can make valid certificates abundant, but it does not automatically create explanations, maintainable libraries, or calibrated human understanding. A mature science of AI-generated mathematics should therefore follow proofs from derivation through presentation and integration to uptake. The multiscale program above turns that philosophical distinction into controlled experiments, longitudinal outcomes, and falsifiable models.

A Reproducibility and computational provenance

The final synthesis script is `code/final_synthesis.py`. It consumes four compact tables: `paired_numina_corrected.csv`, the clean paired `term0.csv`, `matched_hf_records.csv.gz`, and the lean-eval records. The audited term extraction is in `code/paired_term_structure.py`; corrected block and bootstrap routines are in `code/matched_stats.py`. The confirmatory resampling run used 10,000 bootstraps and 10,000 permutations on the CMU ORCHARD cluster, submitted as Slurm job 128378 (completed in 7m58s using eight allocated CPUs; peak recorded RSS 700,868 KiB). Inputs were transferred as a 728 KiB archive with SHA-256 `f779f00a0656a24bf8e7e12c98be6c964fb4071be4d3e88b90b30b62f2f8efeb`. The returned result archive verified against SHA-256 `cac276075ec55b8cfc25ca6cafb91549d8beef824914df5b9a42fdf8203f7f43`.

Bulk corpora and extracted networks remain on remote storage and are not embedded in the repository. This limits third-party end-to-end reproduction. A publication package should deposit immutable paired graph samples, manifests, extraction logs, and licenses in an archival repository.

B Interpretive guardrails

- The paired script sample requires at least one regex-counted premise in each proof and is not the same population as the paired term sample.
- The term sample has high pairwise exclusion due to error-recovery labels. Exclusion can be associated with proof complexity or syntax, so effect sizes may be selected.
- “Human” and “AI” labels are dataset provenance labels, not pure causal treatments; human editing, retrieval, and multi-model pipelines blur the categories.
- Source groups are coarse and only 12 in number. Cluster intervals have limited resolution.
- All term metrics share nodes and edges and are strongly dependent; no multiple-testing narrative should count them as independent replications.
- Tail-index approximations require at least 20 observations above a cutoff; results at different cutoffs have different samples as well as different estimands.
- The belief model is a theoretical cognitive mechanism. Reproducing one published output does not behaviorally validate its update rule, priors, node ontology, or reading process.

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